Shervin Khazaeli

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Portfolio | LinkedIn

Summary

Principal-level AI leader with 10+ years architecting production-grade ML and GenAI systems for critical infrastructure and enterprise clients, and leading the cross-functional teams that build them. Directed teams of data scientists and MLOps/AI engineers while partnering with Product and Engineering to scale AI capabilities from research to production; took systems from pilot to 65+ enterprise clients and $60M+ in operational cost savings. Designed and shipped agentic and GenAI systems (multi-agent orchestration, RAG, LLM-as-judge evaluation, tool-use workflows) grounded in a PhD-level quantitative background in time-series analysis, Bayesian modeling, and probabilistic decision-making. Stays hands-on, independently shipping full-stack AI prototypes from multi-agent pipelines and knowledge graphs to a computer-vision detector trained from scratch. Built enterprise AI governance frameworks from the ground up for auditability, risk management, and regulated-industry readiness.

Core Competencies

Leadership & Strategy: Cross-Functional AI Leadership, Team Building, Hiring & Mentorship, Strategic Planning, Agile Delivery (Scrum, Kanban), Stakeholder Management
AI/ML & GenAI: Agentic AI Systems, Multi-Agent Orchestration (LangGraph), RAG, Knowledge Graphs, LLM-as-Judge Evaluation, Multi-Provider LLMs (Claude, Gemini, OpenAI), Predictive Analytics, Anomaly Detection, Deep Learning, Computer Vision, Reinforcement Learning, Bayesian Modeling
Platform & Architecture: GCP Vertex AI, Kubeflow, BigQuery, Neo4j, PostgreSQL/pgvector, Supabase, FastAPI, Next.js, React, CI/CD, MLOps, Docker
Languages & Frameworks: Python, SQL, TypeScript/JavaScript (Node.js), PyTorch, TensorFlow/Keras, Scikit-learn, LangChain
Governance & Risk: Enterprise AI Governance, Model Cards, NIST AI RMF Alignment, Auditability, Risk

Experience

- **Trinnex (CDM Smith)** — Canada/USA
  Principal / Lead Data Scientist — *2022 – 2026*
  - Architected and scaled an enterprise ML platform (grown from initial pilot) serving 65+ critical infrastructure clients across the U.S., delivering $60M+ in operational cost savings through predictive asset failure modeling, anomaly detection, and infrastructure risk assessment. Built end-to-end in Python on GCP Vertex AI with BigQuery as the primary data layer and Kubeflow-orchestrated, Docker-containerized pipelines; reduced deployment cycle from 4–6 weeks to same-day releases through CI/CD, model versioning, and automated validation.
  - Designed and deployed three production GenAI applications using agentic orchestration (Google Agent Development Kit, Claude Code agents, harness engineering): a RAG-based internal knowledge retrieval system for structured decision support; an AI-powered code analysis agent autonomously reviewing pull requests and generating structured remediation, compressing review cycles from days to hours; and a competitive intelligence platform combining web retrieval, tool use, and structured LLM outputs.
  - Implemented a tri-agent dialectic validation framework (prosecutor / defender / adjudicator) as a production LLM-as-judge evaluation architecture for ML feature validation and data-leakage detection, applying adversarial prompting, function calling, and structured output contracts to automate a review process previously requiring senior data scientist judgment.
  - Directed data scientists and MLOps engineers to design and deploy ML models predicting the material composition of water service lines for 65+ U.S. utilities, directly supporting EPA Lead & Copper Rule Revision (LCRR) compliance; scaled a blockage detector from proof-of-concept to 60+ U.S. sewer sensors.
  - Led a technical team of data scientists and AI engineers, owning hiring and performance reviews and running Agile delivery (Scrum and Kanban) in Azure DevOps and Asana; established end-to-end ML development standards (version control, CI/CD, quality gates, deployment templates, a shared mono-repo library), achieving Google MLOps maturity. Created the "SEAL 6" knowledge-sharing methodology to scale team capabilities and shape the agentic AI product roadmap.
  - Built AI governance into the ML lifecycle of client-facing products (documentation standards, explainability reviews, traceability, validation gates, incident response). Engaged directly with utility officials and infrastructure decision-makers, translating predictive models into operational decisions.

- **Polytechnique Montreal** — Canada
  Research Assistant (Full-time) — *2018 – 2022*
  - Developed and deployed an AI framework for real-time anomaly detection on the Jacques Cartier and (old) Champlain bridges, using enhanced Bayesian dynamic linear models, reinforcement learning, and GPU-accelerated computation; advised the bridges' public operator on structural risk findings.
- **Concordia ECSGA** — Montreal, Canada
  Instructor (Part-time, Contractual) — *2016 – 2018*
  - Designed and taught the graduate course "Probabilistic Graphical Modeling" to 80+ students from all engineering departments: Bayesian networks, probabilistic inference, and sequential models (hidden Markov and state-space models), with hands-on BayesiaLab labs building and evaluating models on real data sets.
- **Sensequake Inc.** — Canada
  Consultant & Coordinator (Part-time, Contractual) — *2016 – 2017*
  - Led client engagements on landmark infrastructure projects including the Thousand Islands Bridge (ON) and Pacific Forestry Centre (BC): installed vibration sensors on site, applied signal processing and data mining to the raw data, and built data-driven seismic assessment models feeding automated client reports.
- **Concordia University** — Canada
  Research Assistant (Full-time) — *2014 – 2017*
  - Built a cloud-based framework on AWS for AI-based structural health monitoring, using signal processing, data mining, and ML classification to detect and localize damage in real time on a simulated multi-story structure.

Independent Projects

- **AI Governance & Readiness Assessment Pipeline** — Independent
  Solo Developer (Prototype) — *2025 – Present*
  - 5-node multi-agent pipeline orchestrated with LangGraph (intake → regulatory research → plan generation → adversarial verification → deliverable generation) that turns a conversational intake into a cited, phased AI-governance roadmap; the adversarial agent re-queries the regulatory corpus to cross-check the plan and can trigger a bounded revision loop. Provider-agnostic LLM layer via LangChain (Gemini, OpenAI, Groq, Mistral) with model-per-node routing and retry/circuit-breaker resilience.
- **Graph-Intelligence Agentic AI Platform for Infrastructure Risk** — Independent
  Solo Developer (Prototype) — *2025 – 2026*
  - "Agents-as-product" architecture on Neo4j: five domain-specialist agents plus an orchestrator that decomposes compound questions, dispatches to specialists in parallel, and synthesizes a single cited answer. Built under a self-imposed Evaluation-Driven Development discipline: every agent shipped with a weighted rubric, a labeled reference set (30%+ adversarial cases), and an LLM-as-judge eval runner gating readiness; all graph access runs through one auditable tool library.
- **Regulatory Precedent Ontology & Knowledge Graph** — Independent
  Solo Developer (Prototype) — *2026*
  - Turned public utility-commission rulings from two U.S. state regulators into a citable precedent knowledge graph: a Pydantic-typed domain ontology loaded into Neo4j, Claude extraction via forced tool use, vector search for passage retrieval, and PageRank over the citation network. Made grounding code-enforced (every fact carries a verbatim quote verified on its page) and set a quality gate before extraction; both compared models failed it, the stronger on a single field, which was then excluded rather than the bar lowered.
- **Forward-Looking Sonar Target Detection & Edge-Readiness Prototype** — Independent
  Solo Developer (Prototype) — *2026*
  - Trained a compact anchor-free (CenterNet-style) object detector from scratch in TensorFlow/Keras on real forward-looking sonar imagery from a public research dataset, then ran INT8 post-training quantization via TFLite with before/after accuracy measurement on a development CPU (mean IoU held at 0.47). Enforced an absolute quality gate after an adversarial review caught a model that passed its tests while predicting a constant box; documented limits in a model card aligned with the NIST AI RMF.

Certifications

- **Artificial Intelligence Governance Professional (AIGP)**, *IAPP* — *2026*
- **Project Management Professional (PMP)**, *PMI* — *2026*
- **Google Project Management Professional**, *Google* — *2025*

Education

- **Polytechnique Montreal** — Canada
  PhD, Civil Engineering (Artificial Intelligence) — *2022*
- **Sharif University of Technology** — Iran
  MSc, Civil Engineering (Applied Mathematics) — *2012*